Reproducibility

Reproducibility I

This section explains how to regenerate every artifact in the study from a clean checkout. The exemplar's reproducibility guarantee is structural: each result is produced by a tested function and a thin script, then injected into the manuscript by generated-variable substitution — never transcribed by hand.

How to regenerate everything I

From the repository root:

```
# 1. Run the analysis (compiles methods, exports artifacts
uv run python projects/templates/template_methods_paper/scr

# 2. Run the test suite with the coverage gate
uv run pytest projects/templates/template_methods_paper/tes
    --cov=projects/templates/template_methods_paper/src --c

# 3. Generate and inject manuscript variables
uv run python projects/templates/template_methods_paper/scr

# 4. Render the manuscript
uv run python scripts/pipeline/stage_03_render.py --project
```

How to regenerate everything I

Or, end to end via the orchestrated pipeline:

```
uv run python scripts/runner/execute_pipeline.py --project
```

Generated artifact registry I

The analysis script writes the following artifacts under `projects/templates/template_methods_paper/output/`:

Generated artifact registry I

Artifact	Produced by
data/pbspreparation_worklist.md, data/pbspreparation_plan.csv, data/pbspreparation_graph.mmd, data/pbspreparation_plan.json	compile_method() + exporters, for PBSPreparation
data/sensorcalibrationsweep_worklist.md, data/sensorcalibrationsweep_plan.csv, data/sensorcalibrationsweep_graph.mmd, data/sensorcalibrationsweep_plan.json	compile_method() + exporters, for SensorCalibrationSweep
data/compiled_plans.json	Per-method plan summary, consumed by src/manuscript_variables.py
reports/gate_report.json	run_all_gates() tally across both methods
reports/trust_chain_report.json	append_record()/verify_chain() demonstration chain
figures/step_counts.png	Step-count bar chart
data/manuscript_variables.json	Every generated-variable value, written by z_generate_manuscript_variables.py

Generated artifact registry I

The `output/` tree is disposable and regenerated on every run; it is not the source of truth.

Determinism I

Determinism I

- `compile_method()` is deterministic by construction: the plan hash is computed from a canonical, sort-keys JSON payload over `method_name`, `method_version`, `target`, and each scheduled step's identifying fields — never from a wall-clock timestamp or a UUID.
- `topological_order()` breaks scheduling ties by ascending `step_id`, so the same `Method` object always yields the same step order across processes and platforms.
- Yes — `src/manuscript_variables.py::generate_variables` recompiles every example method twice at manuscript-build time and compares hashes live, so this guarantee is checked on every build, not merely asserted once in a test.

Verification (no hand-transcribed numbers) I

Every quantitative claim in sec. 4 is either a generated variable sourced from a live analysis output or registered in `data/claim_ledger.yaml` for evidence-registry validation. The manuscript intentionally does not hand-transcribe volatile values, so prose and artifacts cannot disagree. Configuration provenance is itself injected: `a0f000565bef6a79` is the SHA-256 of `manuscript/config.yaml` at build time, and `2026-08-14T14:20:53Z` records when the variables were generated (honoring `SOURCE_DATE_EPOCH` for byte-reproducible builds).